

COM3502/4502/6502 SPEECH PROCESSING

Lecture 6

Frequency Analysis

Time vs. Frequency Domain

- A signal may represent a sequence of *any* kind of measurement
- It is common for signals to represent a sequence of measurements in *time*
- A useful technique for signal analysis is to decompose it into a set of basic pieces that are easier to understand (e.g. *linear trends*)
- Decomposing into *periodic* functions can be thought of as transforming a signal into the '**frequency domain**' (i.e. *repetitions in time can be expressed as a frequency*)

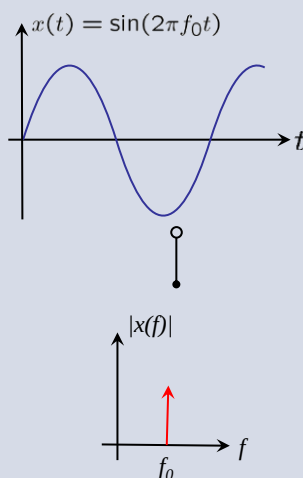
$$f = 1/\tau \text{ cycles-per-second/Hertz}$$

Fourier Analysis



- Introduced by **Joseph Fourier** (1768–1830)
- **Theorem:** any periodic signal of frequency f_0 can be constructed exactly by adding together '**sinusoids**' (*sine waves*) with frequencies ...
 - $f_0, 2f_0, 3f_0, 4f_0, 5f_0$, etc.
- Each sinusoid in this '**Fourier series**' is characterised by its ...
 - frequency
 - amplitude
 - phase
- f_0 is termed the '**fundamental frequency**'
- $2f_0, 3f_0, 4f_0$, etc. are termed the '**harmonics**'

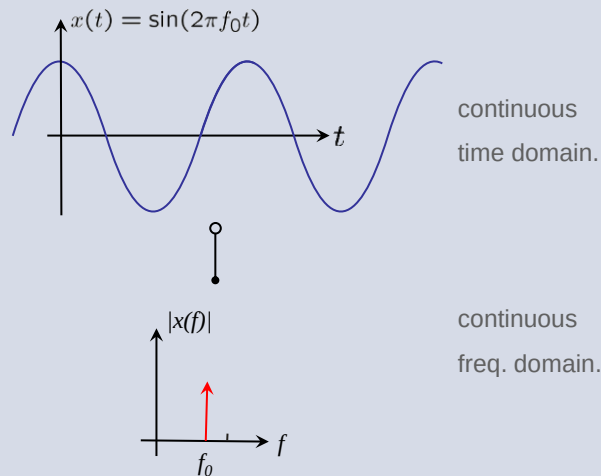
Fourier Analysis - Sinus



continuous
time domain.

continuous
freq. domain.

Fourier Analysis - Cosinus

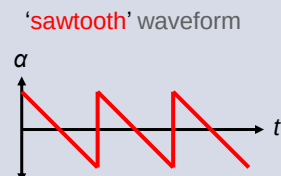


Fourier Analysis - Sawtooth

$$a(t) = \frac{2}{p} \sum_{k=1}^{\infty} \frac{\sin(k\omega t)}{k}$$

$$= \frac{2}{p} \left(\sin(\omega t) + \frac{\sin(2\omega t)}{2} + \frac{\sin(3\omega t)}{3} + \dots \right)$$

All harmonics

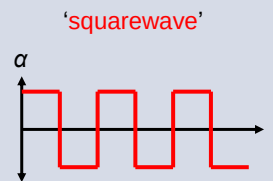


Fourier Analysis – Square Wave

$$a(t) = \frac{4}{p} \sum_{k=1}^{\infty} \frac{\sin((2k-1)\omega t)}{(2k-1)}$$

$$= \frac{4}{p} \left(\sin(\omega t) + \frac{\sin(3\omega t)}{3} + \frac{\sin(5\omega t)}{5} + \dots \right)$$

Odd harmonics

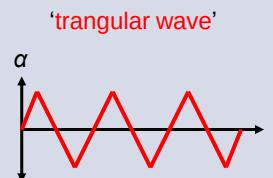


Fourier Analysis – Triangle Wave

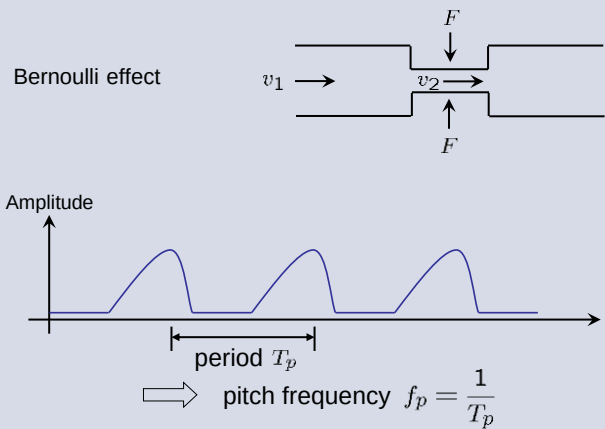
$$a(t) = \frac{8}{p^2} \sum_{k=0}^{\infty} (-1)^k \frac{\sin((2k+1)\omega t)}{(2k+1)^2}$$

$$= \frac{8}{p^2} \left(\sin(\omega t) - \frac{\sin(3\omega t)}{9} + \frac{\sin(5\omega t)}{25} - \dots \right)$$

Odd harmonics
(alternating sign)



Airflow passing the vocal lips

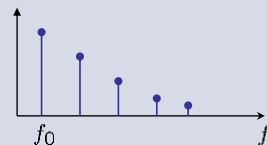


Signals with harmonic structure

Fourier series:

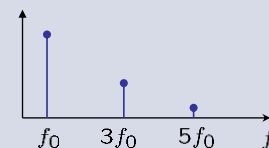
$$f(t) = -\frac{2h}{\pi} \left[\sin(\omega_0 t) - \frac{\sin(2\omega_0 t)}{2} + \frac{\sin(3\omega_0 t)}{3} \mp \dots \right]$$

$$= -\frac{2h}{\pi} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{\sin(k\omega_0 t)}{k}$$



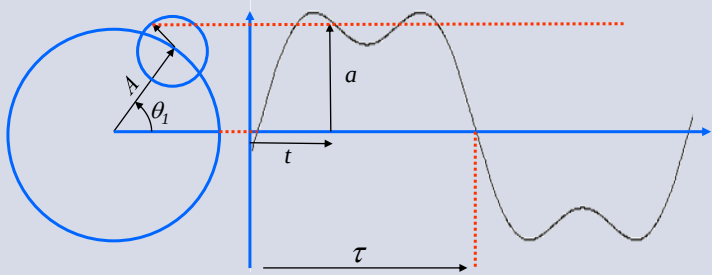
$$f(t) = \frac{8h}{\pi^2} \left[\sin(\omega_0 t) - \frac{1}{32} \sin(3\omega_0 t) + \frac{1}{52} \sin(5\omega_0 t) \mp \dots \right]$$

$$= \frac{8h}{\pi^2} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{\sin((2k-1)\omega_0 t)}{(2k-1)^2}$$



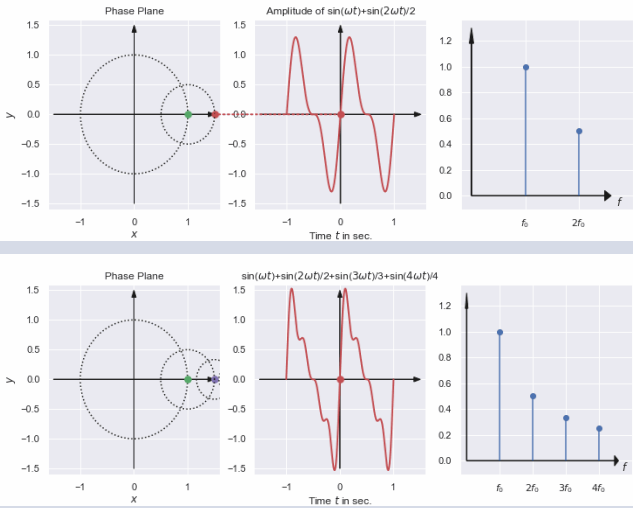
Fourier Analysis

Projection of multiple vectors rotating at constant velocities



- A magnitude of 1st rotating vector
- θ phase angle of 1st vector (radians)
- $f_0 = 1/\tau$ fundamental frequency (cycles per second, Hertz)
- $\omega = 2\pi f_0$ fundamental angular frequency (radians/second)
- $\theta = \omega t$
- $a = A \sin(\omega t) + A/3 \sin(3\omega t)$

Fourier Analysis



The Spectrum

- Any periodic function can be characterised by the amplitude and phase of its sinusoidal components – it's '**spectrum**'
- Periodic signals only have spectral components at *integer multiples* of the fundamental (c.f. *Fourier's theorem*)
- Aperiodic and stochastic signals have spectra that are continuous functions of frequency (i.e. *all possible frequencies*)
- White noise has a *flat* spectrum (i.e. *equal energy at all frequencies*)



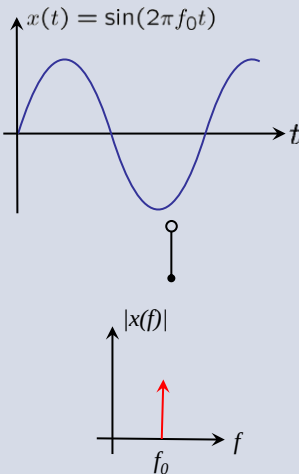
The Power Spectrum

- Previous examples have illustrated the '**power spectrum**', i.e. the energy present at each frequency
- The power of a signal at each frequency is the sum of the squares of its real and imaginary components (e.g. *a sine wave has varying amplitude but constant power*)
- Information (*the phase relationships*) is lost during the calculation of the power spectrum, so this means that the original signal cannot be recovered accurately
- However, a signal's spectrum is fully specified by the magnitude and phase at each frequency ... this is the '**complex spectrum**'

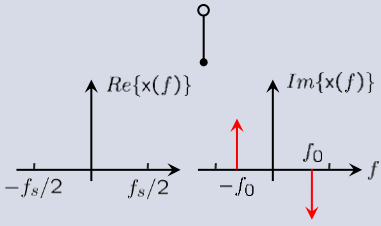
Fourier Analysis - Sinus

continuous
time domain.

continuous
freq. domain.



- Sinus spectrum is imaginary
- Mathematically has also peak at negative frequencies

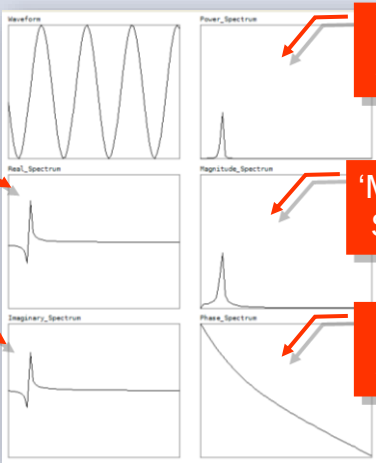


The Complex Spectrum

'Real'
Spectrum

'Imaginary'
Spectrum

Cartesian
coordinates



'Power'
Spectrum

'Magnitude'
Spectrum

'Phase'
Spectrum

Polar
coordinates



The Fourier Transform

- The complex (magnitude and phase) spectrum is obtained from a time signal by means of the '**Fourier transform**'
- No information is lost during the transformation
- This means that the time signal can be recovered from the complex spectrum by means of the '**inverse Fourier transform**'

This slide set has covered ...

- Fourier analysis
- Fundamental frequency
- Harmonics
- The power spectrum
- The complex spectrum
- The Fourier transform
 - (first glance, more in lecture 'Fourier Transform')

Any Questions ?

Raise during lecture or post in the Blackboard Discussion group



Next time ...

The Z-Transform